Denial of Service

The task for the laboratory is to gain practical knowledge on how to perform variable Denial of Service attacks. There is no specific template or structure provided to allow freedom to write essay-type documentation. **This lab must be performed in the Virtual Environment.**

Preparation for the lab:

- Connect to the FTP server using the command **ftp 192.168.69.232** and use **anonymous** credentials.
- Navigate to **ddos** folder using **cd** command.
- Download the **dos_monitor** file using **get** command.
- Exit the ftp utility by typing **exit**
- You have downloaded a file and now you need to make it executable, it can be done using the following command **chmod +x ./dos_monitor**.
- To execute **dos_monitor** you can use the following command **./dos_monitor** (assuming you are in the directory where file is located)

Expect to write at least 500 words (**overall**) discussing the following:

- Is it possible to control CPU utilization?
  - Identify the configuration metric for hping3 ICMP and SYN DoS such that the impact is:
    - 40% CPU Utilization
    - 60% CPU Utilization
    - 85% CPU Utilization
  - Discuss the results and make a note of how much traffic was generated.
- Describe the methodology on how you have identified the number of requests required to have an impact on the HTTP service e.g., through script use.
- Add a discussion on what is the chain of events that have an impact on the computer system. What resources are affected.

You may use diagrams and screenshots to support your discussion.

The planned outline of the lab

You will collaboratively use multiple Denial of Service techniques to create an impact on the HTTP service.

We will be using the following tools **hping3** and **curl** with scripting.

Hping3 – is the utility allows to generate traffic using variety of protocols

sudo hping3 -V -S -p 80 --interval u1000000 192.168.69.209

The parameters:

-V - verbose output i.e. will display what is sent

-S - TCP SYN Mode

-p - Specifying the port

--inerval - is self-explanatory, u represents microseconds, u1000000 - 1 second

192.168.69.209 - Target IP

Using **man** command, identify how to modify the command line to change to ICMP mode.

Curl – is the utility that can perform various HTTP requests

You are given an example bash script which will allow you to make an HTTP request from your kali terminal.

```
#!/bin/bash
for i in {1..2} # {1..2} represents loop iterations i.e. will run 2 times
do
        curl http://192.168.69.209/index.php?encrypt=mystuff > /dev/null 2>&1&
done
```

The >/dev/null 2>&1 means that output from STDOUT and STDERR will not be shown to the user.

To make the requests, you may use any scripting language e.g. Python.

It would be best if you referenced your findings and use of any additional literature.